

ShapeMix-ATAC Simplified Explanation

Contents

1	Bayesian model without fragment length bins	1
2	Bayesian model for ShapeMix	2
3	Example	2

1 Bayesian model without fragment length bins

1.1 Array indices

Index	Meaning
s	which spatial spot?
p	which selected genomic peak?
b	which parent-fragment length bin?
c	which reference cell type?

1.2 Array dimensions

Dimension	Meaning
S	number of spatial spots
P	number of peaks
B	number of fragment length bins=3
C	number of cell types

1.3 Symbols

Symbol	Meaning	Shape	Role
R	reference counts	$C \times P$	fixed
z	effective abundance	$S \times C$	unknown
n	predicted (mean) counts	$S \times P$	calculated
N	observed total counts	$S \times P$	data

1.4 Bayesian model

$$\underbrace{p(z | N)}_{\text{posterior}} \propto \underbrace{p(N | z)}_{\text{likelihood}} \cdot \underbrace{p(z)}_{\text{prior}}$$

The peak count total is modeled by a negative binomial:

$$N \sim \text{NegBinomial}\left(\text{mean } n = z \cdot R, \text{ inverse-dispersion} = \phi_{\text{ref}} \cdot \sum_c z\right)$$

where ϕ_{ref} is a constant. $z \cdot R$ is a dot product between two vectors: for one spot s and one peak p , $n_{s,p} = \sum_c z_{s,c} R_{c,p}$.

Every element of z has the same Gamma prior:

$$z \sim \text{Gamma}(\text{shape} = 2, \text{rate} = 1)$$

This is the frozen version-1 prior. It has mean 2, variance 2, and mode 1.

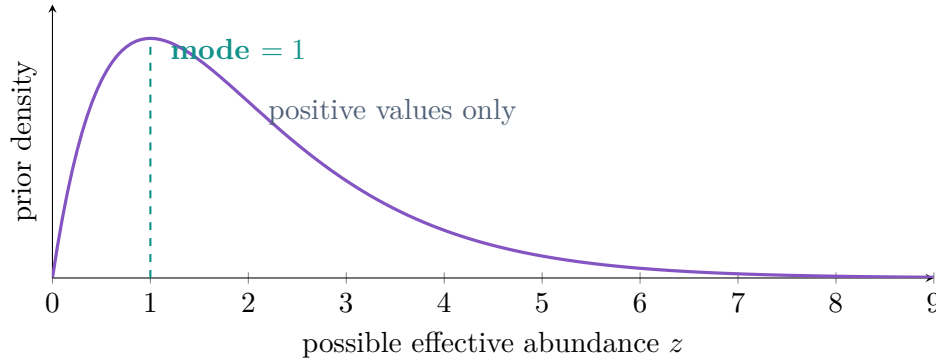


Figure 1: The Gamma(2,1) prior is a probability density over positive effective abundance.

2 Bayesian model for ShapeMix

2.1 Symbols extended to bin counts

Symbol	Meaning	Shape	Role
R_LB	reference rate per length bin	$C \times P \times B$	fixed
n_lb	predicted (mean) bin counts	$S \times P \times B$	calculated
N_LB	observed bin counts	$S \times P \times B$	data

Summing over the three length bins recovers the totals from Section 1:

$$R = \sum_b R_LB, \quad n = \sum_b n_lb, \quad N = \sum_b N_LB.$$

2.2 Bayesian model

$$\underbrace{p(z \mid N_LB, N)}_{\text{posterior}} \propto \underbrace{p(N_LB, N \mid z)}_{\text{likelihood}} \cdot \underbrace{p(z)}_{\text{prior}} = p(N_LB \mid N, z) \cdot p(N \mid z) \cdot p(z)$$

The distribution of the last two terms has already been introduced. For $p(N_LB \mid N, z)$, it is a multinomial distribution:

$$N_LB \mid N \sim \text{Multinomial}(N, \frac{n_lb}{n}) = \text{Multinomial}(N, \frac{z \cdot R_LB}{n})$$

3 Example

3.1 Example without fragment length bins

Assume one spot ($S = 1$), one peak ($P = 1$), and two cell types T and B ($C = 2$), so the abundance vector can be written as $z = (z_T, z_B)$. The fixed reference rates for this peak are:

$$R_T = 12, \quad R_B = 12.$$

Suppose the hidden abundances are:

$$z_T = 1.5, \quad z_B = 0.5.$$

The predicted mean count is the dot product from Section 1:

$$n = z \cdot R = z_T R_T + z_B R_B = 1.5(12) + 0.5(12) = 24.$$

With the example value $\phi_{\text{ref}} = 6$, the inverse-dispersion is $\phi_{\text{ref}} \sum_c z = 6 \times (1.5 + 0.5) = 12$, so the observed total is modeled as:

$$N \sim \text{NegBinomial}(\text{mean} = 24, \text{inverse-dispersion} = 12).$$

A possible observation is $N = 24$. Each abundance also receives its prior score from $z_T, z_B \sim \text{Gamma}(2, 1)$. Because $R_T = R_B$, the swapped mixture $z_T = 0.5, z_B = 1.5$ predicts the same total: $0.5(12) + 1.5(12) = 24$. The total alone cannot distinguish the two recipes; that is what the fragment length bins add.

3.2 Example for the ShapeMix model

Keep the same spot, peak, and mixture. With $B = 3$ length bins (short, mono, long), the binned reference rates split each cell type's peak rate across the bins:

Cell type	R	R_LB (short, mono, long)		
T	12	6	4	2
B	12	2	4	6

Each row sums back to the Section 3.1 rate: $6 + 4 + 2 = 12 = R_T$ and $2 + 4 + 6 = 12 = R_B$, as required by $R = \sum_b R_LB$.

The predicted bin counts add the two cell-type contributions:

$$n_lb = z_T(6, 4, 2) + z_B(2, 4, 6) = (9, 6, 3) + (1, 2, 3) = (10, 8, 6),$$

and their sum recovers the Section 3.1 total: $10 + 8 + 6 = 24 = n$. The predicted bin fractions are:

$$\frac{n_lb}{n} = \left(\frac{10}{24}, \frac{8}{24}, \frac{6}{24} \right).$$

Given the observed total $N = 24$, the bin counts are one multinomial draw:

$$N_LB \mid N = 24 \sim \text{Multinomial}\left(24, \left(\frac{10}{24}, \frac{8}{24}, \frac{6}{24}\right)\right),$$

and a possible observation is $N_LB = (10, 8, 6)$, whose bins sum to the total of the first example: $10 + 8 + 6 = 24 = N$.

The bins are what distinguish the two recipes. The swapped mixture $z_T = 0.5, z_B = 1.5$ predicts the same total 24 but a different split:

$$0.5(6, 4, 2) + 1.5(2, 4, 6) = (3, 2, 1) + (3, 6, 9) = (6, 8, 10).$$

An observed split of $(10, 8, 6)$ therefore favors $z_T = 1.5, z_B = 0.5$.